

Chapter 5: Mathematics Logic

Class 20: Practical / Activities (20 Minutes)

Topic: Pattern Completion Activity, Number Sequence Challenge, Skip Counting Game & Number Puzzle Worksheet

Activity 1: Pattern Completion Activity (5 Minutes)

Objective

To help students identify, understand, and complete different number and shape patterns using logical thinking.

Materials Required

- Pattern Activity Worksheet
- Number Cards
- Shape Cards
- Whiteboard & Marker
- Pencil & Eraser

Instructions

1. The teacher explains that every pattern follows a rule.
2. Display different number and shape patterns on the board.
3. Students identify the rule and complete the missing numbers or shapes.
4. Students explain how they found the answer.
5. The teacher discusses the correct pattern and encourages students to create their own pattern.

Pattern Completion Examples

Number Patterns

1. 5, 10, 15, ____, 25
Answer: 20
2. 2, 4, 6, 8, ____
Answer: 10
3. 100, 90, 80, ____, 60
Answer: 70
4. 12, 24, 36, ____, 60
Answer: 48
5. 50, 100, 150, ____
Answer: 200

Shape Patterns



Answer: 



Answer: 



Answer: 

Teacher Observation Checklist

- ☒ Identifies patterns correctly.
 - ☒ Completes missing numbers.
 - ☒ Explains the pattern rule.
 - ☒ Participates actively.
 - ☒ Demonstrates logical thinking.
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Activity 2: Number Sequence Challenge (5 Minutes)

Objective

To develop students observation, sequencing, and mathematical reasoning skills.

Materials Required

- Number Sequence Cards
- Whiteboard
- Sequence Worksheet
- Timer

Instructions

1. Divide students into small groups.
2. Give each group a number sequence card.
3. Students identify the rule and complete the sequence.
4. The fastest team with correct answers earns points.
5. Discuss different methods used to identify the sequence.

Number Sequence Challenges

1. 10, 20, 30, 40, ____
2. 3, 6, 9, 12, ____
3. 100, 80, 60, 40, ____
4. 7, 14, 21, 28, ____
5. 5, 15, 25, 35, ____

Bonus Challenge

Find the missing numbers:

- 8, __, 24, 32, 40
- __, 50, 75, 100
- 18, 36, __, 72

Teacher Observation Checklist

- ☒ Recognizes sequence rules.
- ☒ Solves sequences accurately.

- ✓ Works cooperatively.
 - ✓ Completes within time.
 - ✓ Participates confidently.
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Activity 3: Skip Counting Game (5 Minutes)

Objective

To improve counting skills, multiplication readiness, and number recognition through fun classroom games.

Materials Required

- Number Flashcards
- Ball (Optional)
- Whiteboard
- Skip Counting Chart

Instructions

1. Students stand in a circle.
2. The teacher announces a skip-counting number.
3. Students take turns saying the next number in the sequence.
4. Any student who gives an incorrect answer sits out for one round.
5. Continue with different skip-counting patterns.

Skip Counting Practice

By 2

2, 4, 6, 8, 10, 12...

By 5

5, 10, 15, 20, 25...

By 10

10, 20, 30, 40, 50...

By 25

25, 50, 75, 100...

By 50

50, 100, 150, 200...

By 100

100, 200, 300, 400...

Fun Challenge

The teacher says:

- "Skip count by 5 starting from 15."
- "Skip count by 10 starting from 40."
- "Skip count by 2 starting from 18."

Students answer one after another without hesitation.

Teacher Observation Checklist

- ☒ Counts accurately.
 - ☒ Responds quickly.
 - ☒ Identifies skip-counting patterns.
 - ☒ Participates enthusiastically.
 - ☒ Demonstrates confidence.
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Activity 4: Number Puzzle Worksheet (5 Minutes)

Objective

To strengthen logical reasoning and mathematical problem-solving through engaging number puzzles.

Materials Required

- Number Puzzle Worksheet
- Pencil
- Whiteboard
- Teacher Answer Key

Instructions

1. Distribute the worksheet to all students.
2. Students solve the puzzles independently.
3. The teacher reviews the answers together.
4. Encourage students to explain their thinking.
5. Appreciate creative problem-solving approaches.

Worksheet Activities

Puzzle 1 – Fill in the Missing Number

1. 6, 12, 18, ____, 30
 2. 9, 18, ____, 36
 3. 20, 40, 60, ____
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Puzzle 2 – Arrange in Ascending Order

45, 18, 62, 29

Answer: 18, 29, 45, 62

Puzzle 3 – Arrange in Descending Order

90, 34, 78, 56

Answer: 90, 78, 56, 34

Puzzle 4 – Even or Odd

Identify whether the following numbers are **Even** or **Odd**:

- 24
- 37
- 60
- 85
- 92

Puzzle 5 – Find the Rule

4, 8, 12, 16, ____

Answer: 20

Teacher Observation Checklist

- ☒ Solves puzzles independently.
- ☒ Applies logical reasoning.
- ☒ Identifies even and odd numbers.
- ☒ Arranges numbers correctly.
- ☒ Completes the worksheet accurately.

Practical Time Distribution

Activity	Duration
Pattern Completion Activity	5 Minutes
Number Sequence Challenge	5 Minutes
Skip Counting Game	5 Minutes
Number Puzzle Worksheet	5 Minutes
Total Practical Time	20 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Identify and complete **number and shape patterns** using logical reasoning.
- Recognize and extend **number sequences** by applying mathematical rules.
- Perform **skip counting** accurately using different intervals such as 2, 5, 10, 25, 50, and 100.
- Solve **number puzzles** involving missing numbers, ascending and descending order, and even and odd numbers.
- Develop stronger **observation, critical thinking, mathematical reasoning, and problem-solving skills** through interactive activities.
- Build confidence in applying mathematics concepts to classroom tasks and everyday situations.